

ABSTRACT

In urban waste management, traditional dustbins pose challenges such as inefficient waste collection, overflow issues, and unhygienic handling. Existing systems lack automation, leading to delayed disposal, manual intervention, and increased labor costs. Additionally, municipal authorities struggle with ineffective monitoring of dustbin fill levels, resulting in untimely waste clearance and environmental pollution. The absence of real-time tracking and notification mechanisms further exacerbates the inefficiencies in waste collection and management.

To overcome these drawbacks, this project proposes the development of a Smart Dustbin with automatic lid opening, fill level monitoring, location detection, and real-time notification to mobile devices. The system integrates IoT sensors to detect the dustbin's fill level and transmits data to a cloud-based platform. When the bin reaches its capacity, a notification is sent to the relevant authorities via a mobile application, ensuring timely waste collection. GPS technology is incorporated to track the location of filled bins, optimizing waste collection routes. The automation of the lid reduces direct human contact, promoting hygiene. This solution enhances waste management efficiency, minimizes environmental hazards, and contributes to the development of smart cities.

CONTENTS

1. INTRODUCTION	1
2. TECHNOLOGY IN GENERAL	2
2.1 Block Diagram	2
2.2 Block Diagram Explanation	3
2.2.1 Automatic Lid Opening and Waste Segregation	3
2.2.1.1 Microcontroller (Arduino Uno)	3
2.2.1.2 Ultrasonic Sensor	4
2.2.1.3 Soil Moisture Sensor	4
2.2.1.4 Inductive Proximity Sensor	5
2.2.1.5 IR Sensor	5
2.2.1.6 MG 995 Servo motor	5
2.2.1.7 SG90 Servo motor	6
2.2.2 Fill Level Monitoring	6
2.2.2.1 Ultrasonic sensor	6
2.2.2.2 GPS Module	6
3. TECHNOLOGY IN SPECIFIC	7
3.1 Arduino uno	7
3.2 Ultrasonic sensor	8
3.3 IR sensor	9
3.4 Soil Moisture Sensor	9
3.5 Inductive proximity Sensor	10
3.6 Servo motor (MG995)	10
3.7 Servo motor (SG90)	11

3.8 GPS Module	CONTENTS	12
4. HARDWARE IMPLEMENTATION		14
4.1 Schematic diagram		14
4.1.1 Schematic diagram of Automatic Lid Opening and Waste Segregation		14
4.1.2 Schematic diagram of Fill Level Monitoring and GPS		15
4.2 Working		16
4.2.1 Working of Automatic Lid Opening and Waste Segregation		16
4.2.2 Working of Fill Level Monitoring and GPS		17
4.3 List of Components		18
5. SOFTWARE IMPLEMENTATION		19
5.1 Algorithm		19
5.1.1 Automatic Lid Opening and Waste Segregation		19
5.1.2 Fill Level Monitoring and GPS		20
5.2 Program		21
5.2.1 Automatic Lid Opening and Waste Segregation		21
5.2.2 Fill Level Monitoring and GPS		26
6. RESULTS		30
7. CONCLUSION		31
8. REFERENCES		32
9. APPENDIX		33

LIST OF FIGURES

1. Fig 1.1 Automatic lid open and waste segregation
2. Fig 1.2 Fill Level Monitoring
3. Fig 2.1 Arduino UNO
4. Fig 2.2 Ultrasonic Sensor
5. Fig 2.3 Soil Moisture Sensor
6. Fig 2.4 Inductive Proximity Sensor
7. Fig 2.5 IR Sensor
8. Fig 2.6 MG995 Servo Motor
9. Fig 2.7 SG90 Servo Motor
10. Fig 2.8 Ultrasonic sensor
11. Fig 2.9 GPS Module
12. Fig 3.1 Arduino UNO pin mapping
13. Fig 3.2 ATmega328P pin mapping
14. Fig 3.3 Ultrasonic Sensor pinout
15. Fig 3.4 IR Sensor pinout
16. Fig 3.5 Soil Moisture Sensor pinout
17. Fig 3.6 Inductive Proximity Sensor
18. Fig 3.7 Servo Motor (MG995) pinout
19. Fig 3.8 Servo Motor (SG90)
20. Fig 3.9 Servo Motor (SG90) pinout
21. Fig 3.10 uBlox NEO-6 pinout
22. Fig 3.11 GPS module
23. Fig 3.12 BLYNK output 1
24. Fig 3.13 BLYNK output 2
25. Fig 4.1 Schematic diagram of Automatic Lid Opening and waste segregation
26. Fig 4.2 Schematic diagram of Fill Level Monitoring and GPS
27. Fig 6.1 Output

CHAPTER 1

INTRODUCTION

Nowadays efficient waste management has become a critical challenge for both households and urban communities. The increasing volume of waste, coupled with the need for proper segregation and disposal, demands innovative solutions to promote sustainability and hygiene. Enviro guard :A smart waste management system provides efficient and sustainable solutions for the above mentioned problems regarding waste disposal.

EnviroGuard has an automatic lid-opening mechanism, which eliminates the need for physical contact, enhancing convenience and hygiene. The ultrasonic sensor in the dustbin detects a user's presence and opens its lid effortlessly, reducing the risk of contamination due to bins being kept open over a long period of time. Additionally, the system incorporates real-time fill level monitoring through the Blynk IoT application, allowing users to track waste accumulation . This feature ensures timely waste collection, preventing overflow and maintaining cleanliness.

EnviroGuard addresses one of the biggest challenges in waste management—automated segregation. Using a combination of sensors in the input bin, the system categorizes waste into dry, wet, and metal categories before directing them into separate compartments below. This makes recycling easier and minimizes manual sorting efforts, contributing to a greener environment.

By adopting EnviroGuard, households, offices, and public spaces can take a significant step toward responsible waste disposal, reducing environmental impact while embracing the convenience of modern automation. This smart dustbin is more than just a container—it's a guardian of cleanliness and sustainability in the digital age.

CHAPTER 2

TECHNOLOGY IN GENERAL

2.1 BLOCK DIAGRAM

AUTOMATIC LID OPENING AND WASTE SEGREGATION

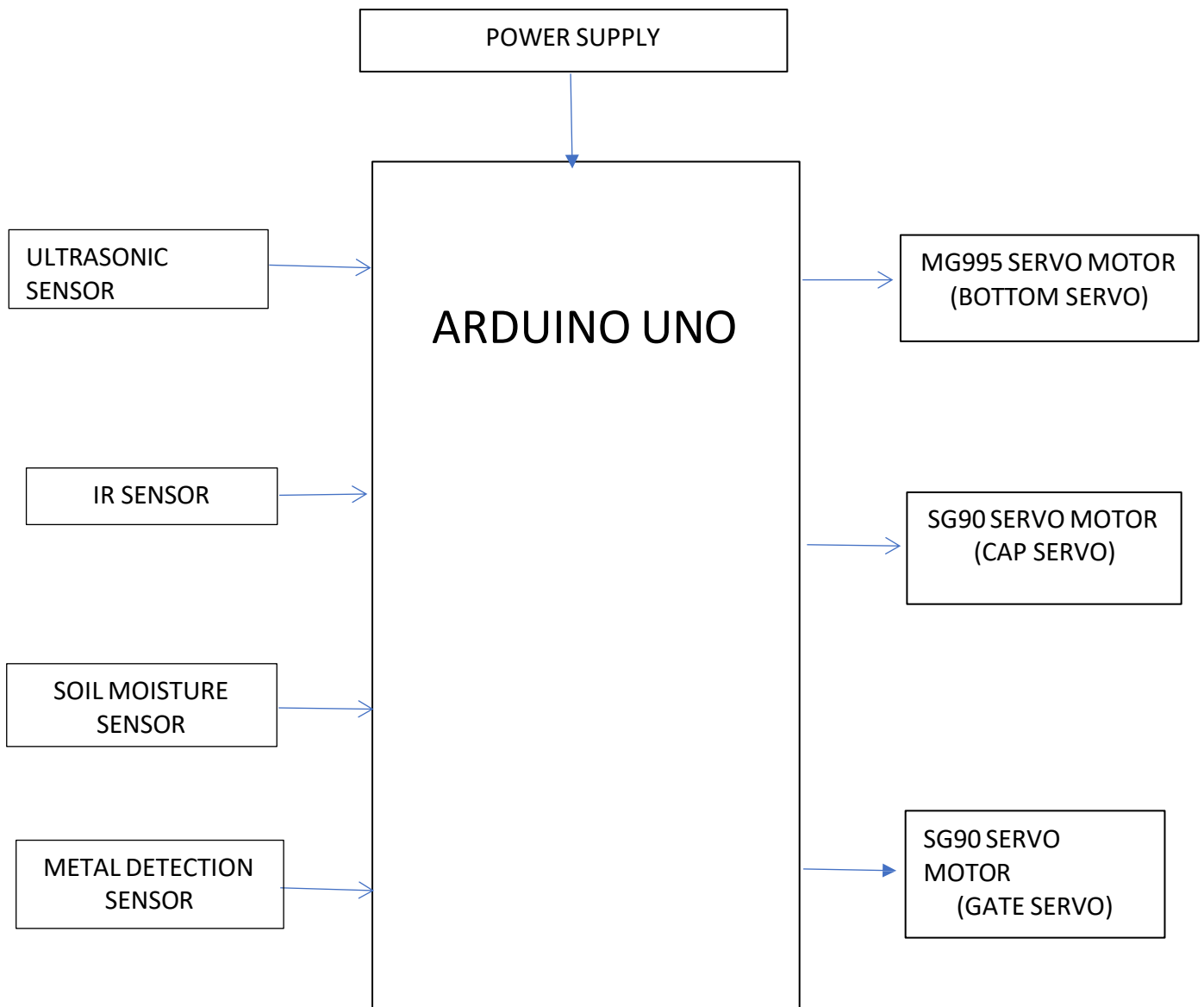


Fig1.1

Block diagram of Automatic Lid Opening and Waste Segregation

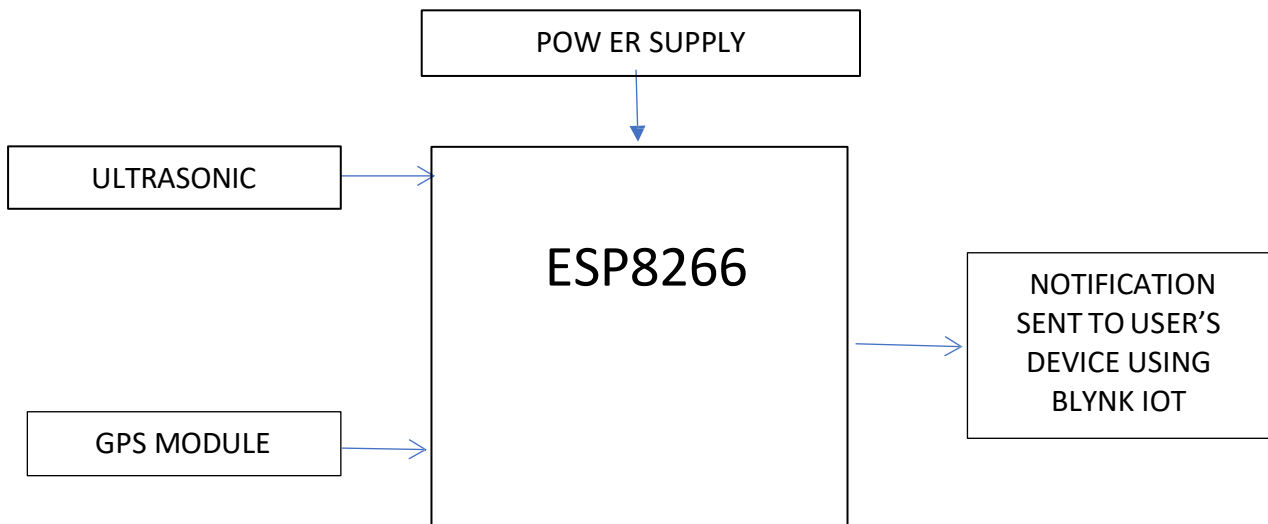
FILL LEVEL MONITORING AND GPS

Fig 1.2

Block diagram of Fill Level Monitoring and GPS

2.2 BLOCK DIAGRAM EXPLANATION**2.2.1 AUTOMATIC LID OPENING AND WASTE SEGREGATION****2.2.1.1 Microcontroller (Arduino UNO)**

In the smart waste bin project, the Arduino Uno acts as the main controller. It powers and reads data from the ultrasonic sensor to measure the bin's fill level, controls the servo motor to open the lid automatically, processes the data, and communicates with the ESP8266 to send bin status updates to the Blynk IoT platform. It serves as the central unit managing all sensor inputs and outputs.



Fig 2.1
Arduino Uno

2.2.1.2 Ultrasonic sensor

The ultrasonic sensor detects if a person is present in front of the bin depending on the threshold value that is set and if yes; then the lid opens automatically.



Fig 2.2
Ultrasonic Sensor

2.2.1.3 Soil Moisture Sensor

A threshold value for the soil moisture is set, below which the waste input is classified as wet waste.

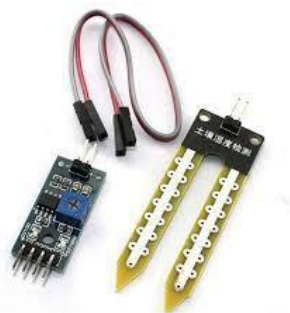


Fig 2.3
Soil Moisture Sensor

2.2.1.4 Inductive proximity sensor

The inductive proximity sensor is used for the detection metal waste in waste segregation.



Fig 2.4

Inductive Proximity Sensor

2.2.1.5 IR Sensor

The IR sensor is used to detect if any waste has been input into the bin. If a waste is detected by the IR sensor yet has not been classified into either wet or metal waste, it is classified as dry waste.



Fig 2.5

IR Sensor

2.2.1.6 MG995 Servo motor

The servo motor placed at the bottom moves the base on which the bins are placed and the corresponding bin of the waste detected is move and placed under the input bin.



Fig 2.6

Servo Motor – MG995

2.2.1.7 SG90 Servo motor

The two of the SG90 servo motors used, one of which is for the automatic lid opening and the other placed at the base of the input bin. The bin automatically opens when the ultrasonic sensor detects the presence of a person in front of the bin. The base of the input bin is opened after the corresponding waste bin of the waste input has reached below it.



Fig 2.7

Servo Motor – SG90

2.2.2 FILL LEVEL MONITORING

2.2.2.1 Ultrasonic sensor

The two ultrasonic sensor placed on the walls of the bin above the metal bin and the wet waste bin measures the fill level of two bins at a time.



Fig 2.8

Ultrasonic Sensor

2.2.2.2 GPS module

The GPS module is used to track the location of the bins that are full. The GPS module sends the latitude and longitude of the bin and the type of waste bin that is full. The Blynk IOT app has been used to display the levels of the bins and also the latitude and longitude of the bins.



Fig 2.9

GPS Module

CHAPTER 3

TECHNOLOGY IN SPECIFIC

3.1 MICROCONTROLLER (ARDUINO UNO)

Arduino Uno utilizes an 8-bit ATmega328P microcontroller quite effectively on its board mostly for various electronic projects. It comprises other components like crystal oscillator and voltage regulator alongside ATmega328P and has serial communication capabilities too. Supporting microcontroller functionality heavily relies on supplementary circuitry underneath it. Arduino Uno have 14 digital I/O pins with 6 capable of PWM output and features 6 analog inputs alongside a USB connection a power jack an ICSP header and reset button.

In this project, the Arduino Uno serves as the main controller when using a basic ESP8266 module which is used for WiFi communication. The Arduino Uno reads data from the ultrasonic sensor, controls the servo motor for the bin lid, and sends the data to the ESP8266 to update the Blynk IoT platform

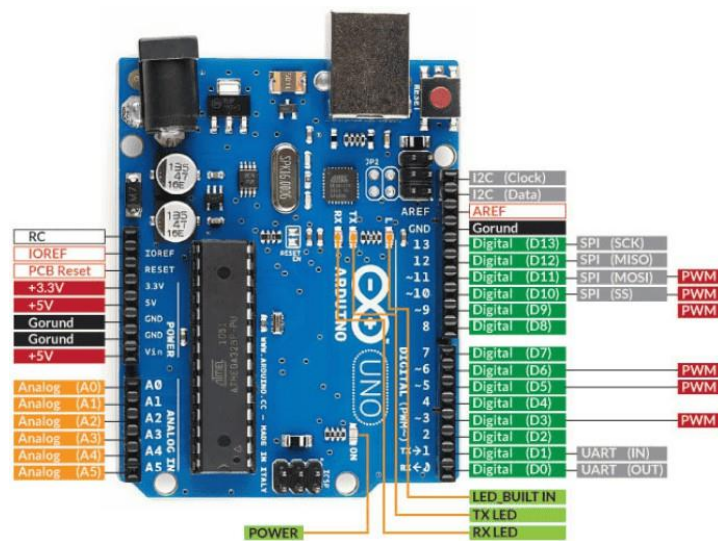


Fig 3.1

Arduino Uno Pinout



Fig 3.2

ATmega328P pin mapping

3.2 ULTRASONIC SENSOR

A device utilizing ultrasonic waves measures distance or detects objects pretty accurately with high frequency sound waves emitted suddenly. Sensors emit high-frequency sound waves usually around 20 kHz and 1 MHz and measure time taken for echo return after bouncing off objects. Time-of-flight measurement enables sensor calculation of object distance with remarkably high precision fairly accurately every single time. Ultrasonic sensors are widely utilized owing largely to ability providing remarkably accurate measurements sans physical contact with an object.

There are a total of 3 ultrasonic sensors used in which one of it is used for the purpose of automatic lid opening and the other two for fill level monitoring. It works by emitting ultrasonic waves and measuring the time it takes for the echo to bounce back from the surface of the waste. Based on this time, the sensor calculates the distance between the sensor and the top of the waste. This distance is then used to determine how full the bin is. As the level of waste increases, the measured distance decreases, allowing the system to update the bin level in percentage and trigger alerts when the bin is full.



Fig 3.3

Ultrasonic Sensor Pinout

3.3 IR SENSOR

IR Sensor Module has built-in IR transmitter and IR receiver sending out infrared light and detecting reflected infrared radiation from obstacles nearby. It finds obstacles and facilitates short-range and somewhat medium-range communication pretty effectively. Onboard a potentiometer exists for adjusting detection range somewhat crudely. IR sensor transmits digital data in form of infrared light. Obstacle detection LED indicator sits squarely on module board glowing brightly. LED turns ON upon sensor getting logical 1 and goes OFF when it gets logical 0 essentially signaling a binary state. IR Sensor detects presence of waste in input bin and waste not classified as wet or metal waste gets classified as dry waste.2

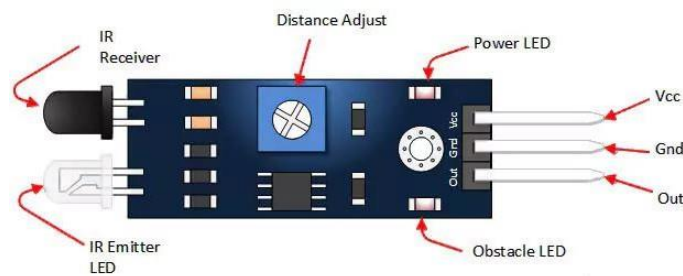


Fig 3.4
IR Sensor Pinout

3.4 SOIL MOISTURE SENSOR

A soil moisture sensor comprises two conducting plates functioning somewhat like a probe and serving as a rather variable resistor together somehow. Volumetric water content gets measured quite accurately by them underground. Two probes facilitate current passage through soil thereby yielding resistance values for quantifying moisture levels effectively underground. Resistance decreases markedly as sensor plates get submerged in water thereby getting better conductivity. Soil conductivity plummets drastically in arid conditions so with less water soil conducts measurably less electricity hence exhibiting higher resistance. In this project soil moisture sensor is used in waste segregation part to detect wet waste. This is done by first setting a threshold value. The condition that is set is that if the input value drops below the threshold value then the waste is classified as wet waste.

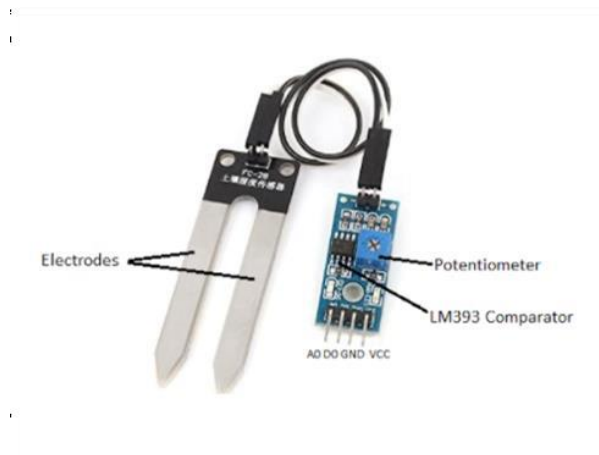


Fig 3.5

Soil Moisture Sensor Pinout

3.5 INDUCTIVE PROXIMITY SENSOR

The SN04-N is a NPN NO (Normally Open) inductive proximity sensor that detects the presence of metallic objects within a 5mm range. It works by generating an electromagnetic field and detecting changes in the field when a metal object approaches. This sensor is commonly used in automation and industrial applications for tasks like metal detection, collision detection, and limit switch applications.



Fig 3.6

Inductive Proximity Sensor

3.6 SERVO MOTOR(MG995)

The MG995 servo motor operates using Pulse Width Modulation (PWM) to control its position. A PWM signal is sent to the motor's control wire, where the width of the pulse determines the angle of rotation. Typically, a 1 ms pulse corresponds to about 0°, a 1.5 ms pulse to 90°, and a 2 ms pulse to 180°, with these pulses repeating every 20 milliseconds. Inside the servo, a small DC motor is connected to a set of gears and a potentiometer. The

potentiometer continuously monitors the shaft's position and sends feedback to an internal control circuit. This circuit compares the desired position (based on the PWM signal) with the actual position, and then drives the motor accordingly to move the shaft to the correct angle. This feedback mechanism ensures precise and stable positioning.

In this project, the servo motor(bottom servo) is used to rotate the base on which the bins are placed in such a way that the corresponding bin under the input bin is of the type of waste detected in the input bin. The servo motor is at an initial position of 90 degree with the dry waste bin under the input bin.

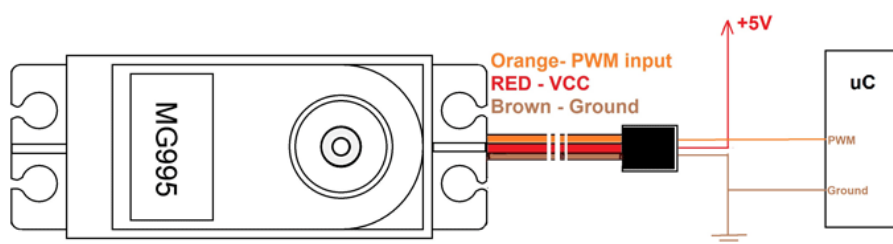


Fig 3.7

Servo Motor – MG995 Pinout

3.7 SERVO MOTOR(SG90)

The SG90 servo motor also operates using Pulse Width Modulation (PWM) to control its shaft position. A PWM signal is sent to the control wire, where the duration of the pulse determines the angular position of the motor. Typically, a pulse width of 1 ms moves the shaft to around 0°, 1.5 ms to 90°, and 2 ms to 180°, with the signal repeating every 20 milliseconds. Internally, the SG90 contains a small DC motor, a set of plastic gears, and a potentiometer attached to the output shaft. The potentiometer continuously measures the shaft's current position and provides feedback to the internal control circuit. This circuit compares the actual position with the target position set by the PWM signal and adjusts the motor's rotation until the shaft reaches the desired angle. This closed-loop control system allows the SG90 to achieve accurate and stable positioning within its ~180° range.

The SG90 servo motor is used for automatic lid opening. When the ultrasonic sensor detects a person in front of the bin, the servo motor rotates 180 degrees and opens the lid of the bin automatically.

After the type of the waste has been detected and the corresponding waste bin has been placed under the input bin by the MG995 servo motor, the SG90 servo motor at the base of

the input bin rotates 180 degrees and opens the base and the waste falls into its corresponding bin. After a delay of 5ms the base of the input bin closes and after a delay of 2ms the lid of the bin closes automatically.

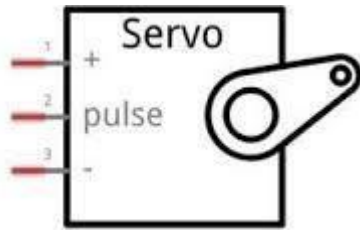


Fig 3.8

Servo Motor SG90



Fig 3.9

Servo Motor Pinout

3.8 GPS MODULE

The NEO-6M GPS module operates by communicating with multiple GPS satellites to determine its geographic location. It receives signals from at least four satellites, each transmitting data about its position and the exact time the signal was sent. By calculating the time delay between transmission and reception for each signal, the module uses trilateration to compute its position in terms of latitude, longitude, and altitude. The NEO-6M contains a u-blox 6 GPS chip and outputs location data in the standard NMEA format via a serial interface, which can be read by microcontrollers like Arduino or Raspberry Pi. It also includes an onboard EEPROM to store configuration settings and a ceramic antenna or SMA connector for reliable signal reception. Due to its high accuracy and fast signal acquisition, the NEO-6M is widely used in tracking systems, navigation, and timing applications.

To prevent overflowing of waste and to maintain a sanitary condition, when a particular bin is full its location is sent to the users phone which has been simulated using the Blynk Iot application. A notification including the type of the waste bin, the longitude and latitude is received by the user.

13	GND	GND	12
14	MOSI/CFG_COM0	RF_IN	11
15	MISO/CFG_COM1	GND	10
16	CFG_GPS0/SCK	VCC_RF	9
17	Reserved	Reserved	8
NEO-6 Top View			
18	SDA2	VDDUSB	7
19	SCL2	USB_DP	6
20	TxD1	USB_DM	5
21	RxD1	EXTINT0	4
22	V_BCKP	TIMEPULSE	3
23	VCC	SS_N	2
24	GND	Reserved	1

uBlox NEO-6 Pinout

Fig 3.10

uBlox NEO-6 Pinout

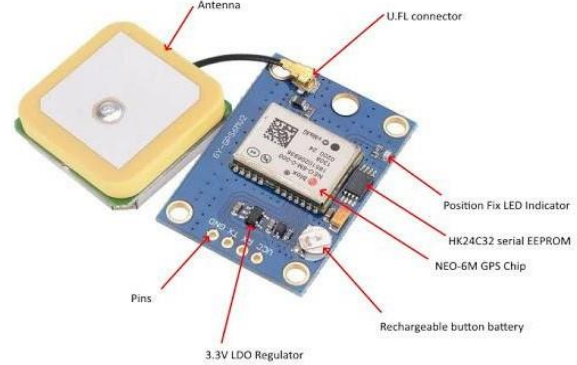


Fig 3.11

GPS Module Pinout



Fig 3.12

BLYNK output 1

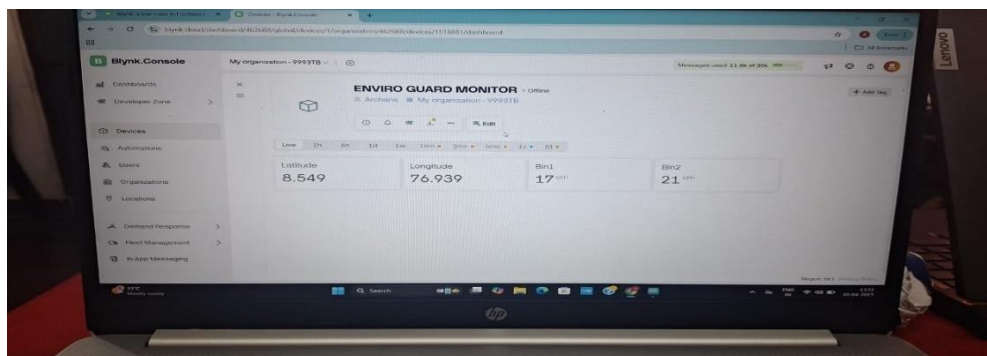


Fig 3.13

BLYNK output 2

CHAPTER 4

HARDWARE IMPLEMENTATION

4.1. SCHEMATIC DIAGRAM

4.1.1 Automatic lid opening and waste segregation

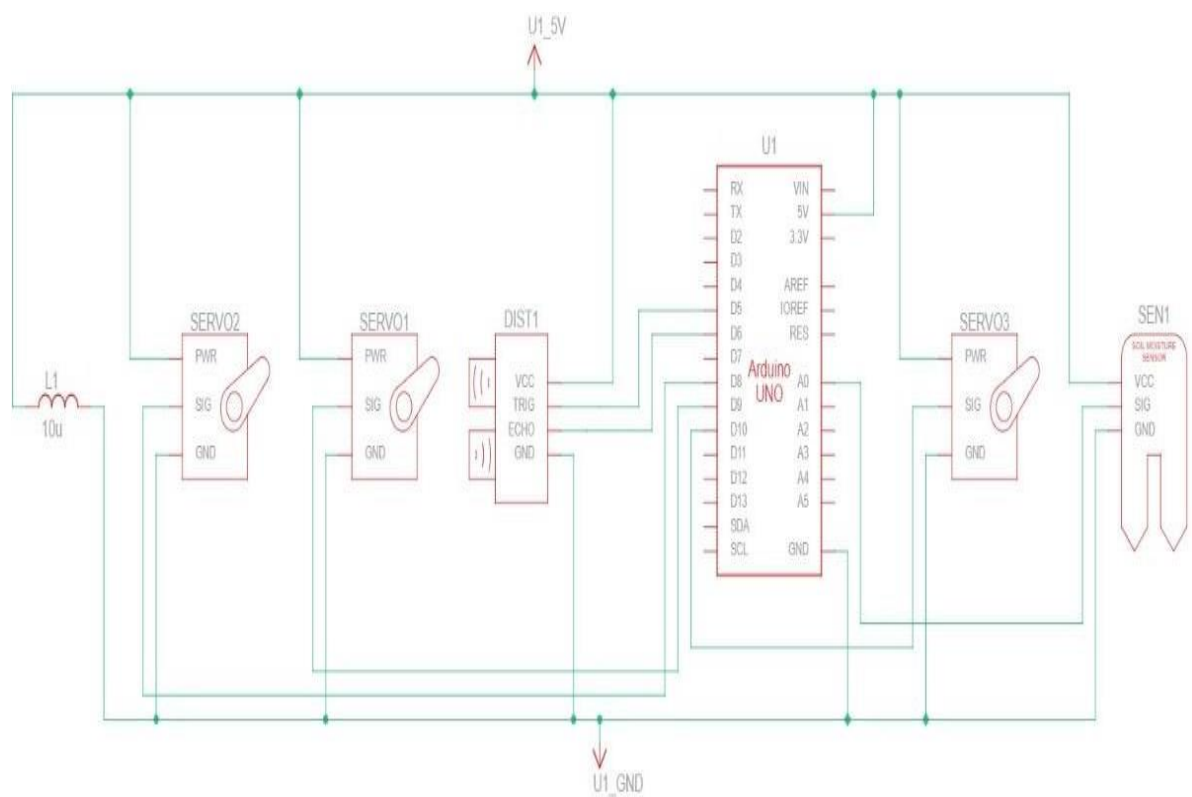


Fig 4.1

Schematic diagram of Automatic Lid Opening and Waste Segregation

4.1.2 Fill level monitoring and GPS location tracking

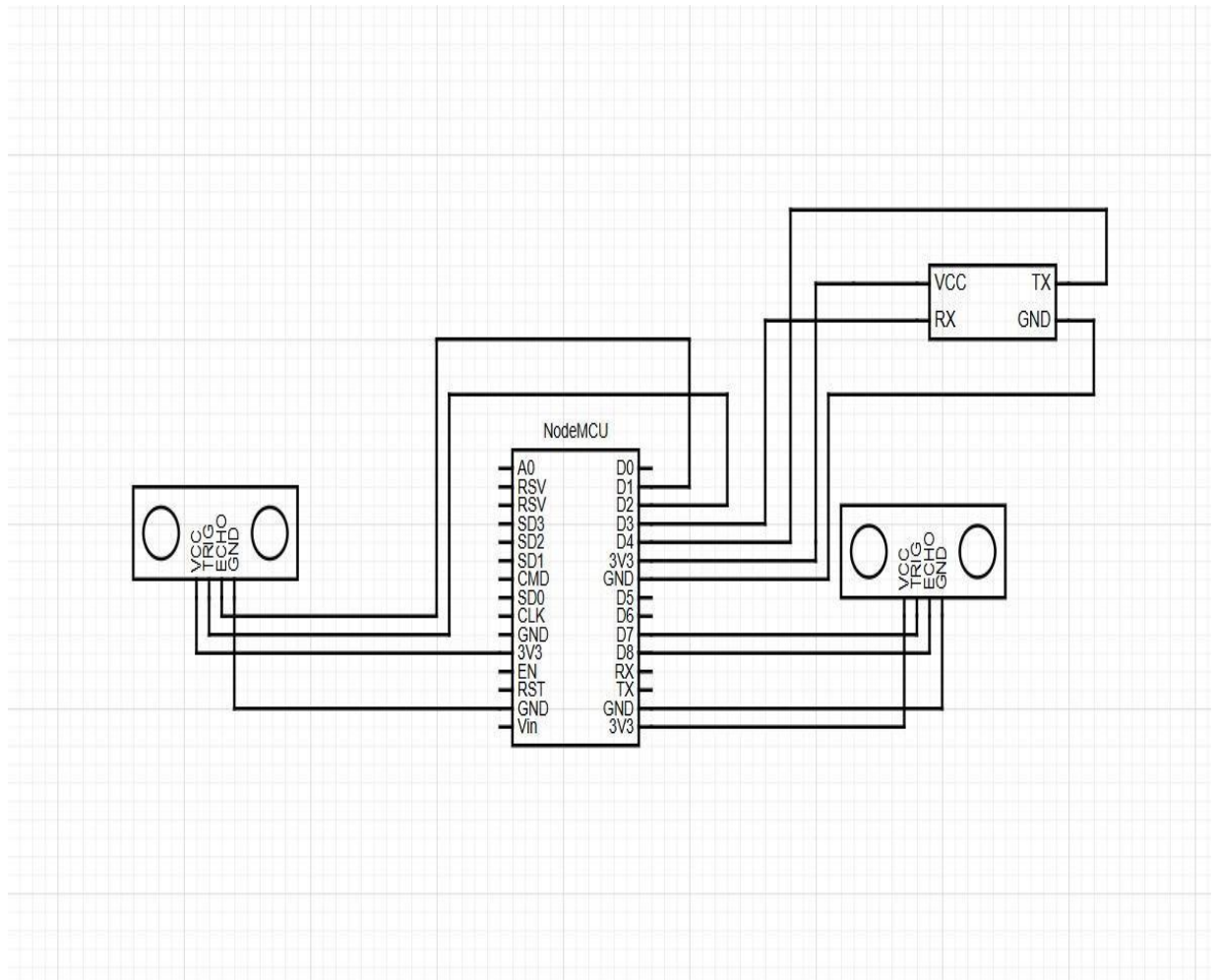


Fig 4.2

Schematic diagram of Fill level monitoring and GPS location tracking

4.2 WORKING

4.2.1 AUTOMATIC LID OPENING AND WASTE SEGREGATION

Enviro guard: A smart waste management system is an automated system that segregates the waste without human contact so that it will promote improved hygiene and better waste management. When the human is detected near the bin, the system starts operation and the ultrasonic sensor is used to detect presence of human. The lid of the bin is automatically opened by a servo motor to enable a user to dispose of waste without touching the bin, when a person is detected within a distance threshold of 10 cm. Upon opening the lid, an IR sensor checks for the presence of waste. In case of waste detection, the system classifies it through the use of a moisture sensor and a metal sensor. Wet waste is thus classified as the waste if the moisture level is below a specified threshold of 750. If the metal sensor is present and there is metal detected and the moisture level is above the threshold, it is considered metal waste. If not, the waste is considered dry. After the type of waste is determined, the internal platform is rotated by a bottom servo motor at 0 degrees for wet, 180 degrees for metal, and 90 degrees for dry. On being positioned the input bin opens by rotating another servo to 90 degrees so that the waste can fall into the appropriate compartment. Then the gate closes and system is reset to its initial state waiting for the next user. By utilizing this smart waste bin, automatic waste segregation in domestic and public area is proved to be a feasible and sanitary way.

4.2.2 FILL LEVEL MONITORING AND GPS TRACKING

The proposed system attempts to measure three types of wastes, dry metal and wet using ultrasonic sensors and tracks it in real-time through a GPS module. For this purpose, one ultrasonic sensor is for metal bin and the other for dry and wet bin that measures the distance between the sensor and the surface of the waste inside the bin. It is supposed to work on measuring the emission of ultrasonic waves and time calculation for their echo return. GPIO pins for dry bin are connected while metal and wet bins are connected through different GPIO channels on ESP8266 also known as NodeMCU microcontroller. It keeps connecting SoftwareSerial by attaching it with GPS module reading latitude-longitude continuously. In case there is some temporary unavailability of GPS data, it has already set default coordinates so location tracking won't be interrupted. It transmits all sensor data relevant to waste levels along with their current location via virtual pins to Blynk IoT platform by connecting to Wi-Fi. Since this information gets updated once every two seconds, remote monitoring gets possible through Blynk mobile app in real-time. within this system an alert mechanism has also been embedded identifying when threshold distances predefined have been crossed by any of the bins towards fullness. Informative message-contains such information as bin type along with its located GPS coordinates will be generated once a bin is full Thus offers smart efficient solutions toward monitoring levels within waste along with location tracking proper urban management.

4.3 LIST OF COMPONENTS

1. Arduino UNO
2. ESP8266(NODE MCU)
3. IR sensor
4. Inductive proximity sensor
5. Soil moisture sensor
6. Ultrasonic sensor (3)
7. MG995 Servo Motor
8. SG90 Servo motor (2)
9. GPS module
10. 12 V Power Supply

CHAPTER 5

SOFTWARE IMPLEMENTATION**5.1 ALGORITHM****5.1.1 ALGORITHM FOR AUTOMATIC LID OPENING AND WASTE
SEGREGATION**

1. First, power the system by activating the communication bus (servo motors and sensors) and configuring the pins. Position all servos to their starting positions (bin centered, gate closed, cap closed).
2. Check for the presence of a human within a 10 cm threshold by using an ultrasonic sensor to measuring the distance.
3. Open cap servo to allow for waste insertion (while a human is detected) and delay for 4 seconds.
4. The waste will be checked for by an IR sensor (active LOW). Stop the process of opening cap with no waste detection; close the cap if it is opened without any waste.
5. If waste exists, read data from moisture and metal sensors to classify:
 - If sensor reading is below 750, consider “wet” and turn the bottom servo to 0°.
 - If metal sensor value is high, consider “metal” and turn the bottom servo to 180°.
 - Otherwise, assume to be “dry” and set the servo to 90°.
6. After classification, bottom platform should be solid for 5 seconds. Then with gate servo at 90°, the waste is dropped into the corresponding bin.
7. After 5 seconds, return the gate servo's position to 0 degrees, waiting for 2 seconds before continuing with the detection cycle.
8. If no humans are detected in range, pause the timeline to recheck after a short amount of time.

5.1.2 ALGORITHM FOR FILL LEVEL MONITORING AND GPS MONITOR

1. Initialization: The system start by initializing the ESP8266 WiFi module with the selected network credentials and the Blynk authentication token. Serial communication is set up for debugging and for receiving GPS data. The pins of the ultrasonic sensor for the Dry and Metal waste bins are set to output (trigger) and input (echo).
2. Timer Setup: A timer is configured to call the sendData(func) every 2 seconds to collect and send data periodically.
3. Distance measurement: In the sendData(function) the system sends the ultrasonic sensors to measure the distance inside the Dry and Metal bins; the echo time is then converted to distance using the speed of sound.
4. Data transfer to Blynk: the measured distances will be transmitted by Virtual Pins V2 and V3 to the Blynk app, allowing you to remotely monitor bin levels.
5. GPS Location Handling The system reads data from the GPS module continuously. If a valid GPS location can be determined it takes the current latitude and longitude coordinates. Otherwise it returns to default hardcoded coordinates.
6. Basically to send the GPS data to Blynk: The latitude and longitude (real or fallback) data is sent to the Blynk app via virtual pins V0 and V1.
7. Logic: The system is checking the status of the bin. Dry bin is considered full when the distance from it is less than or equal to 8 cm. Metal bin is considered full when the distance from it is between 5 cm and 14 cm. If either of the above conditions are met then an alert message is generated which states which bin(s) are full and also the current location.
8. Alert: the generated alert message is printed to the Serial Monitor for debugging; there's an option (currently commented out) to send this alert as a log event to Blynk.

9. Loop Function: The loop() function runs Blynk. run(30) and timer. run(30) continuously to maintain connection and enable timely execution of the sensor reading / data transmission tasks.

5.2 PROGRAM

5.2.1 CODE FOR AUTOMATIC LID OPENING AND WASTE

SEGREGATION

```
#include <Servo.h>

// Servo objects
Servo capServo;
Servo bottomServo;
Servo gateServo;

// Pin Definitions
const int capServoPin = 11;
const int bottomServoPin = 10;
const int gateServoPin = 9;

const int trigPin = 7;
const int echoPin = 8;

const int moisturePin = A0;
const int metalPin = 2;
const int irpin = 4;
// Thresholds
const int distanceThreshold = 10;
const int moistureThreshold = 750;
bool waste_flag = false;
```

```
long duration;
int distance;
void setup() {
    Serial.begin(9600);

    capServo.attach(capServoPin);
    bottomServo.attach(bottomServoPin);
    gateServo.attach(gateServoPin);

    pinMode(trigPin, OUTPUT);
    pinMode(echoPin, INPUT);
    pinMode(metalPin, INPUT);
    pinMode(irpin, INPUT);

    capServo.write(0);    // Cap closed
    bottomServo.write(90); // Default bin
    gateServo.write(0);   // Gate closed

    Serial.println("System ready...");
}

void loop() {
    distance = getDistance();

    Serial.print("Distance: ");
    Serial.println(distance);

    if (distance <= distanceThreshold) {
        Serial.println("Human detected!");

        capServo.write(180);
```

```
delay(4000);
```

```
//capServo.write(0);
```

```
//delay(1000);
```

```
int irVal = digitalRead(irpin);
```

```
int moistureVal = analogRead(moisturePin);
```

```
int metalVal = digitalRead(metalPin);
```

```
Serial.print("Moisture: ");
```

```
Serial.println(moistureVal);
```

```
Serial.print("Metal: ");
```

```
Serial.println(metalVal);
```

```
String wasteType = "none";
```

```
bool waste_flag = false;
```

```
if (irVal == LOW) { // IR detected waste (assuming active LOW)
```

```
    if (moistureVal < moistureThreshold) {
```

```
        wasteType = "wet";
```

```
        rotateBottomServo(0);
```

```
        waste_flag = true;
```

```
    } else if (metalVal == 1) {
```

```
        wasteType = "metal";
```

```
        rotateBottomServo(180);
```

```
        waste_flag = true;
```

```
    } else {
```

```
        wasteType = "dry";
```

```
        rotateBottomServo(90);
```

```
        waste_flag = true;
```

```
    }  
    } else {  
        Serial.println("IR: No waste detected.");  
    }  
    Serial.print("Detected Waste Type: ");  
    Serial.println(wasteType);  
    if (waste_flag == true) {  
        delay(5000);  
        rotateGateServo(90);  
        delay(5000);  
        rotateGateServo(0);  
        //waste_flag = false;  
        //delay(3000);  
    }  
    delay(2000);  
} else {  
    Serial.println("No Human detected!");  
    capServo.write(0);  
    delay(500);  
}  
}
```

// Get distance from ultrasonic sensor

```
int getDistance() {  
    digitalWrite(trigPin, LOW);  
    delayMicroseconds(2);  
    digitalWrite(trigPin, HIGH);  
    delayMicroseconds(10);  
    digitalWrite(trigPin, LOW);
```

```
    duration = pulseIn(echoPin, HIGH);
```

```
    return duration * 0.034 / 2;
}

// Smooth servo movement for platform
void rotateBottomServo(int targetPos) {
    int currentPos = bottomServo.read();
    if (currentPos < targetPos) {
        for (int pos = currentPos; pos <= targetPos; pos++) {
            bottomServo.write(pos);
            delay(10);
        }
    } else {
        for (int pos = currentPos; pos >= targetPos; pos--) {
            bottomServo.write(pos);
            delay(10);
        }
    }
}

// Smooth servo movement for gate
void rotateGateServo(int targetPos) {
    int currentPos = gateServo.read();
    if (currentPos < targetPos) {
        for (int pos = currentPos; pos <= targetPos; pos++) {
            gateServo.write(pos);
            delay(10);
        }
    } else {
        for (int pos = currentPos; pos >= targetPos; pos--) {
            gateServo.write(pos);
            delay(10);
        }
    }
}
```

5.2.2 CODE FOR FILL LEVEL MONITORING AND LOCATION TRACKING

```
#define BLYNK_TEMPLATE_ID "TMPL3JbtwuVjf"
#define BLYNK_TEMPLATE_NAME "ENVIRO GUARD MONITOR"
#define BLYNK_AUTH_TOKEN "u[REDACTED]"
#define BLYNK_PRINT Serial
#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
#include <SoftwareSerial.h>
#include <TinyGPS++.h>

char auth[] = BLYNK_AUTH_TOKEN;

char ssid[] = "C[REDACTED]"; // Replace with your WiFi name
char pass[] = "[REDACTED]" // Replace with your WiFi password

// Pins for Dry Waste
const int trigDry = 12;
const int echoDry = 14;

// Wet Waste
//const int trigWet = 15;
//const int echoWet = 13;

// Metal Waste
const int trigMetal = 0;
const int echoMetal = 2;

#define GPS_RX 13
#define GPS_TX 15
```

```
#define GPS_BAUD 9600
//define sound velocity in cm/uS
#define SOUND_VELOCITY 0.034
#define CM_TO_INCH 0.393701
#define BIN_HEIGHT 15.0 // in cm
#define FULL_THRESHOLD 5.0

long durationDry, durationMetal;
float distanceDry, distanceMetal;
//float distanceInch;
TinyGPSPlus gps;
SoftwareSerial gpsSerial(GPS_RX, GPS_TX); // GPS TX → D7 (RX)

BlynkTimer timer;
float latitude, longitude;

void setup() {
  Serial.begin(115200);
  Blynk.begin(auth, ssid, pass);

  // Call sendSensor() every 2 seconds
  timer.setInterval(2000L, sendData);
  pinMode(trigDry, OUTPUT);
  pinMode(echoDry, INPUT);

  //pinMode(trigWet, OUTPUT);
  //pinMode(echoWet, INPUT);

  gpsSerial.begin(GPS_BAUD);
  Serial.println("System ready");

  pinMode(trigMetal, OUTPUT);
  pinMode(echoMetal, INPUT);
}
```

```
int getDistance(int trigPin, int echoPin) {
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);

    long duration = pulseIn(echoPin, HIGH);
    return duration * SOUND_VELOCITY / 2;
}

void sendData() {
    distanceDry = getDistance(trigDry, echoDry);
    distanceMetal = getDistance(trigMetal, echoMetal);

    Blynk.virtualWrite(V2, distanceDry);
    Blynk.virtualWrite(V3, distanceMetal);

    Serial.print("Dry Bin Distance (cm): ");
    Serial.println(distanceDry);
    Serial.print("Metal Bin Distance (cm): ");
    Serial.println(distanceMetal);

    // --- GPS LOCATION ---
    while (gpsSerial.available()) {
        gps.encode(gpsSerial.read());
    }

    if (gps.location.isValid()) {
        latitude = gps.location.lat();
        longitude = gps.location.lng();
    } else {

        latitude = 8.549058;
```



```
longitude = 76.938528;
Serial.println("Using hardcoded coordinates...");
}

Blynk.virtualWrite(V0, String(latitude, 6)); // Latitude
Blynk.virtualWrite(V1, String(longitude, 6)); // Longitude

Serial.print("Latitude: ");
Serial.println(latitude, 6);
Serial.print("Longitude: ");
Serial.println(longitude, 6);

// Alert logic
bool dryFull = distanceDry <= 8;
bool metalFull = (distanceMetal > 5.0 && distanceMetal < 14.0);

if (dryFull || metalFull) {
  String alertMsg = "Full Bin Alert: ";
  if (dryFull) alertMsg += "Dry ";
  if (metalFull) alertMsg += "Metal ";
  // Append location info
  alertMsg += "| Location: ";
  alertMsg += String(latitude, 6);
  alertMsg += ", ";
  alertMsg += String(longitude, 6);
  //Blynk.logEvent("alert", alertMsg);
  Serial.println("Alert Sent: " + alertMsg);
}
Serial.println("-----");
}

void loop() {
  Blynk.run();
  timer.run();
}
```

CHAPTER 6

RESULT

In this project, we successfully built a smart waste management system that can automatically separate and monitor waste. The system is able to sort waste into three types: **dry, wet, and metal**. To do this, we used a **soil moisture sensor** to detect wet and dry wastes, a **proximity sensor** to identify metal waste. The separation worked well and helped reduce the need for manual sorting.

We also added an **automatic lid opening** mechanism. This makes it easy and clean to use, without having to touch anything.

To avoid overflowing, we included a fill level monitoring system that checks how full the bin is. This helps make sure the bin gets emptied on time.

The system also has a **GPS module (u-blox NEO-6M)** that sends the location of the bin. This can help when tracking bins in different areas or planning collection routes.

We connected everything using the **ESP8266 Wi-Fi module** and linked it to the **Blynk app**, which shows bin status, location, and alerts when the bin is full. The app was easy to use and worked well in our testing.

Overall, the system worked as expected. This project shows that smart waste management using sensors and IoT can make waste collection efficient and eco-friendly.

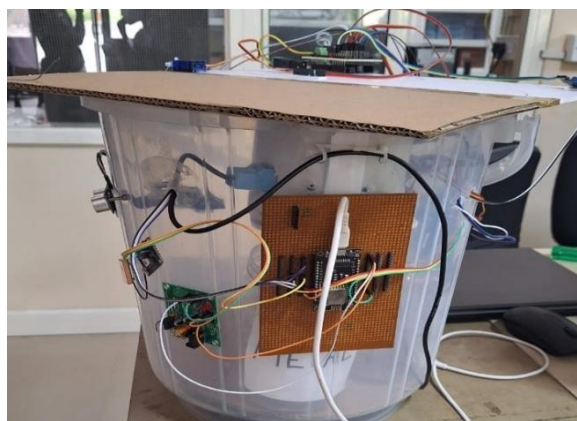


Fig 6.1

Output

CHAPTER 7

CONCLUSION

The ENVIRO GUARD -A SMART WASTE MANAGEMENT SYSTEM project presents a modern and efficient solution for urban waste management by integrating smart technologies into traditional waste bins. With features such as automatic lid opening, fill level monitoring, location detection, and real-time notifications to mobile devices, the system enhances hygiene, user convenience, and operational efficiency. Automatic lid opening reduces contact, promoting better sanitation, while fill level monitoring ensures timely waste collection and prevents overflow. Location tracking enables accurate bin placement and management, and real-time notifications keep both users and waste management authorities informed. Overall, bin contributes to creating cleaner, smarter, and more sustainable cities through innovative IoT-based waste handling.

REFERENCES

- [1] Abd Manaf L, Abu Samah MA, Mohd Zukki NI. Municipal solid waste management in Malaysia: Practices and challenges 2009;29[11]:2902–6
- [2] Thompson A, Afolayan A, Ibidunmoye E, editors. Application of Geographic Information System to Solid Waste Management [PDF Download Available]. 2013 Pan African International Conference on Information Science, Computing and Telecommunications [PACT]; 2013; Lusaka, Zambia: IEEE
- [3] Waste Management and It's Challenges in Malaysia [press release]. 2016-03-20 2016
- [4] <https://justdoelectronics.com/smart-dustbin-with-esp8266-and-gps>
- [5] lastminuteengineers.com
- [6] randomnerdtutorials.com